Indian Geography

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Energy Resources

Coal

Petroleum

Uranium

Thorium

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Indian coal

- Gondwana formation (later than Carboniferous time)
- Inferior
- Cocking coal... X
- Metallurgical coal...X
- Important for electricity generation
- Volatile, Sulphur + moisture is high
- Calorific value is low

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Coal: Bituminous

States	Mineral rich regions
Jharkhand	Damodar valley [Bokaro, Jharia, Dhanbad, Giridih, Daltonganj]
WB	Extension of Damodar valley (raniganj and Barakar formation) Birbhum, Darjeeling (Pankhwali coal field, Darlin coalfield) Burdwan, Bankura
Odisha	Mahanadi basin Talcher valley (Talcher thermal plant)
Chhatisgarh	Narmada, Son rift valley
MP	Chindwara Jhilmil (cocking coal to Bhilai steel plant)
AP	Singreni coal field (Ramadugam Thermal plant) Kotagudam coal field
MH	Nagpur-kampti coal field

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Coal: Lignite

states	Mineral rich regions
Assam	Makum
TN	Neyvelli
RJ	Palana
GJ	Umarsar (Kutchh)

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Coal: Anthracite & Peat

states	Mineral rich regions
Anthracite	
J&K	Kalakot
Peat	
KR	Mangrove-Karee soil
WB	Gangetic Delta

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Problems of Indian coal

- Not enough high grade cocking coal – depend on import
- Good and low grade coal found together – selective mining = wasteful
- Washing, dressing, blending, briquetting of inferior coal needed
- Washing – lower down ash content – waste removed – low weight – transportation cost reduced

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Problems of Indian coal

- Open-caste mining – smuggling, air pollution
- Illegal mining – ecologically dangerous
- Deep mining techniques primitive = high casualty
- Coal field contains Coal Bed Methane
- CBM policy for exploration and production in 1997

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Problems of Indian coal

- Policy does not allow simultaneous exploration of coal and CBM
- CBM exploration and production allowed only in pure seam gas bearing blocks
- CBM blocks and coal blocks are separately auctioned by the govt.
- Coal mines with Methane are dangerous for coal mining – allowed to escape= wasteful

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Problems of Indian coal

- Poor connectivity from mines to consumer locations
- Efficiency of Indian coal mining is very low – lack of tech + equipment
- Output per man shift = India(3), world avg. (7)
- Coal import is increasing despite large reserve

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Underground coal- Gassification

- Underground coal gasification is a method to extract gas from deep, unrecoverable coal reserves, where manual (labour) mining is impossible or costly.
- They dig two wells
  - injection well: water+ oxygen+ gasification agent pumped from here
  - production well: synthetic gas (syngas) comes out from here

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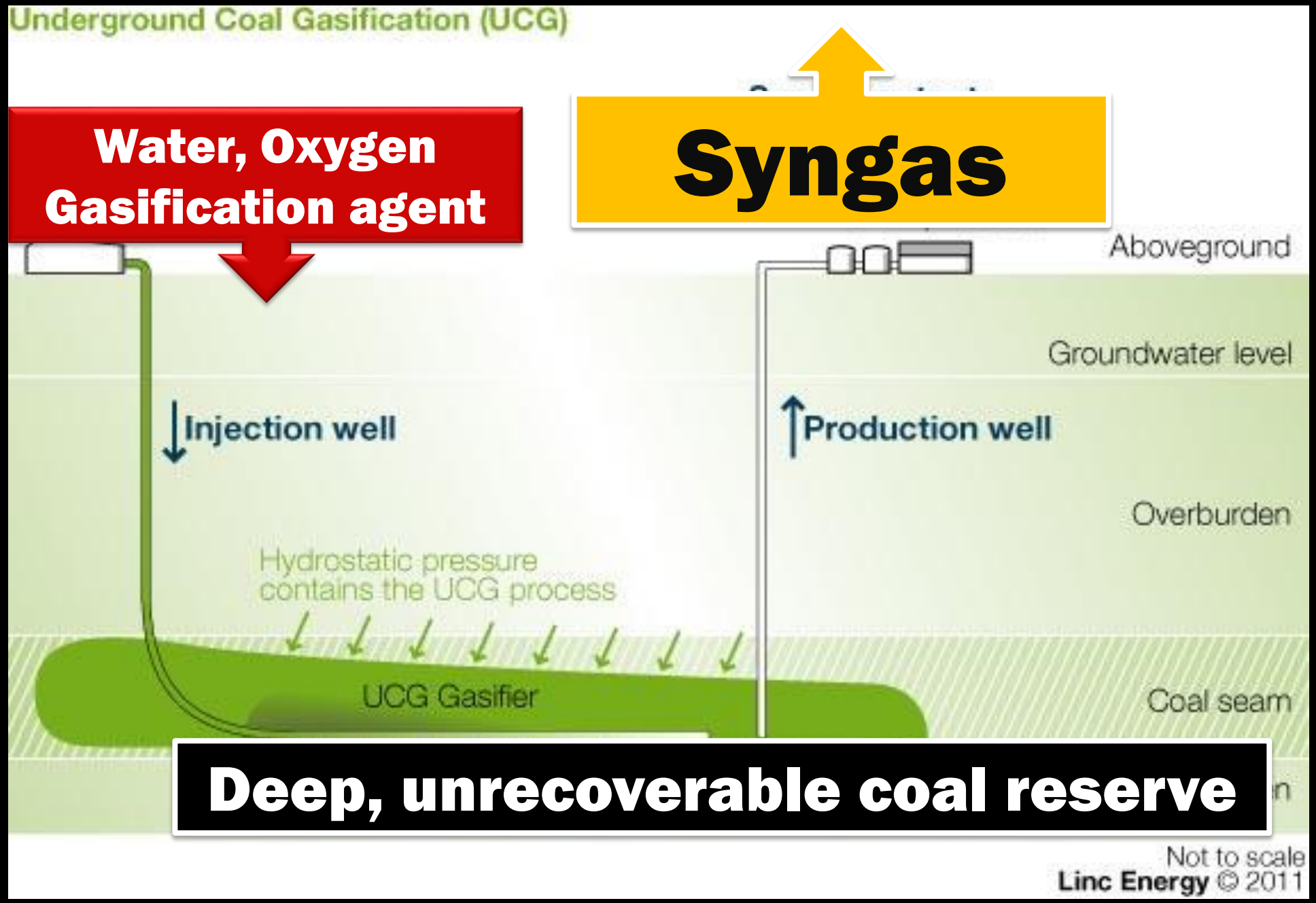
Underground Coal Gasification (UCG)
Water, Oxygen
Gasification agent
Syngas
Aboveground
Groundwater level
Injection well
Production well
Overburden
Hydrostatic pressure
contains the UCG process
UCG Gasifier
Coal seam
Deep, unrecoverable coal reserve
Not to scale
Linc Energy © 2011

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Underground coal- Gasification: Benefits

1) Syngas can be used in generating electricity, manufacturing of hydrogen & fertilizers.
2) Sulphur, mercury, arsenic, tar, ash etc. by-products remain underground = less pollution.
3) needs less water than conventional mining and hydraulic fracking

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Energy minerals

~~Coal~~

Petroleum

Uranium

Thorium

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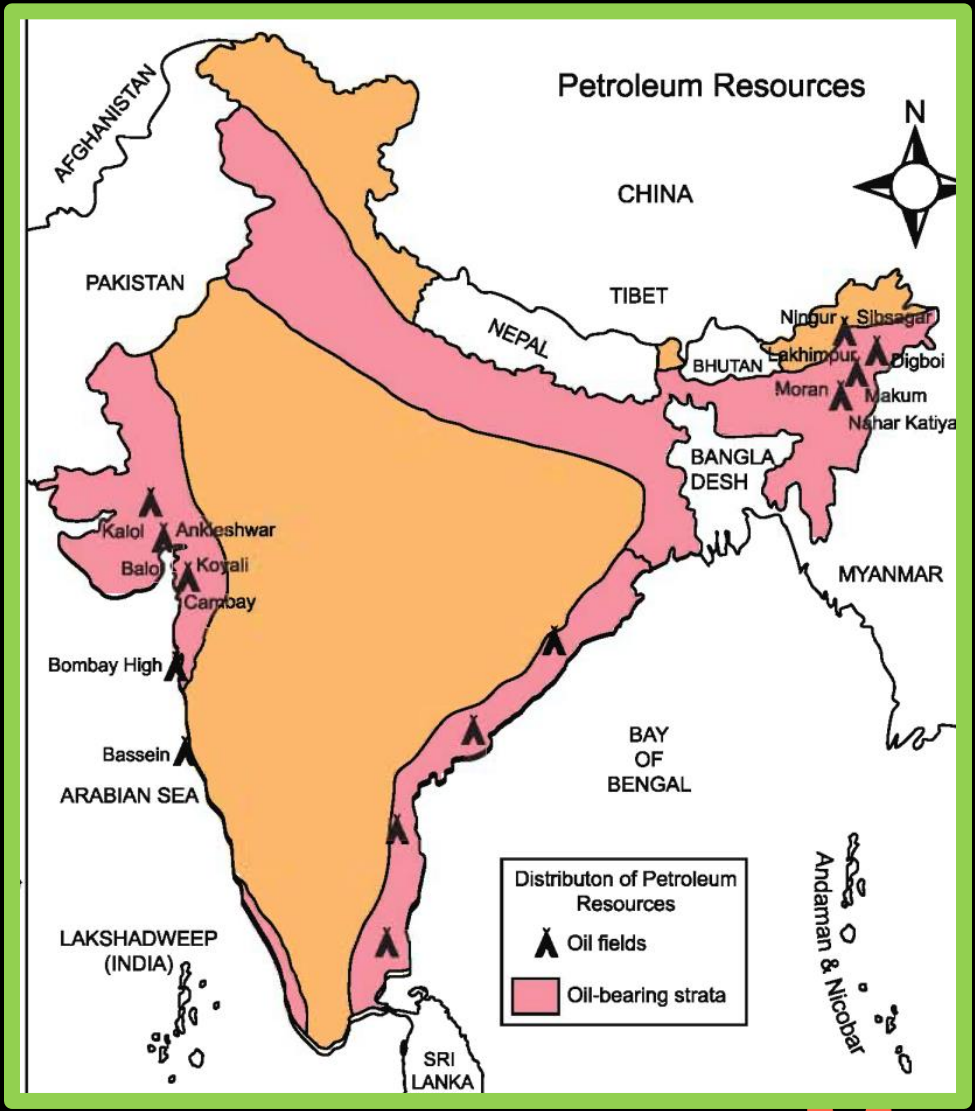
Petroleum Reserves



1) Saurashtra region (GJ, RJ, gulf of Khambhat, Off-shore Mumbai)
2) Upper Assam-Shillong shelf
3) Cuttak shelf
4) KG Basin

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Petroleum

states	Mineral rich regions	Refineries
Assam	Digboi, Neharkatia, Sibsanagar, Dibrugarh	Digboi, Noonmati, Barauni, Bongaigaon
GJ	On-shore: Kheda, Mehsana Off-shore: Gandhar oil filed Aliya bet	Sanand refinery Hazira, Koyali refianries
RJ	Barmer (by Cairns)	

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Petroleum

States	Mineral rich regions
MH	Nilam oil field (Bombay High) – 60% of India's production South Bassain (larger reserve than Nilam)
KG Basin	Rawa oil field (By Reliance and Niko) Off-shore Kaveri basin- Narimanam oil field

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New Petroleum exploration

- Cairn Energy, which bought a block in Rajasthan in 2002 and soon struck oil. The find in Barmer was far from the sea. The crude is waxy, and difficult to transport.
- Now it is planning to lay a heated pipeline to Salaya and export the crude.
- Reliance extracting oil in Krishna-Godavari basin offshore, found gas as well.

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Petroleum refining industry

- One of the 8 core industries of India
- India import crude oil
- 1st refinery in India: Digboi in Assam
- 2nd Tarapur, in 1954
- India exports refined petroleum products 10% of the production

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Petroleum

REFINARIES

Public sector	17
Private	3
JV	2
Total	22

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Petroleum

PSU (17)

PSU Refineries		
1	Guwahati	Indian Oil Corporation Limited
2	Barauni	
3	Koyali	
4	Haldia	
5	Mathura	
6	Digboi	
7	Panipat	
8	Bongaigaon	
9	Mumbai	Hindustan Petroleum Corporation Limited
10	Visakhapatnam	
11	Mumbai	Bharat Petroleum Corporation Limited
12	Kochi	
13	Manali	Chennai Petroleum Corporation Limited
14	Nagapattinam	
15	Numaligarh	Numaligarh Refinery Ltd.
16	Mangalore	MRPL
17	Tatipaka, AP	ONGC

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Refinaries

Private (3)

1. **Reliance**
   1. Jamnagar, GJ
   2. Yanam, Puducherry
2. **Essar**
   1. Vadinar, GJ
3. **Cairn**
   1. Amalpuram, AP

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1. Bina (MP): Bharat + Oman
2. Bathinda: HPCL + Mittal

Refinaries

JV (2)

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Find correct statements

A. India's refining capacity exceeds demand.

B. India has only 3 private sector refineries

C. Both A and B

D. Neither A nor B

Ans. C Both correct

1. Skip 2. Attempt 3. Mark n Review

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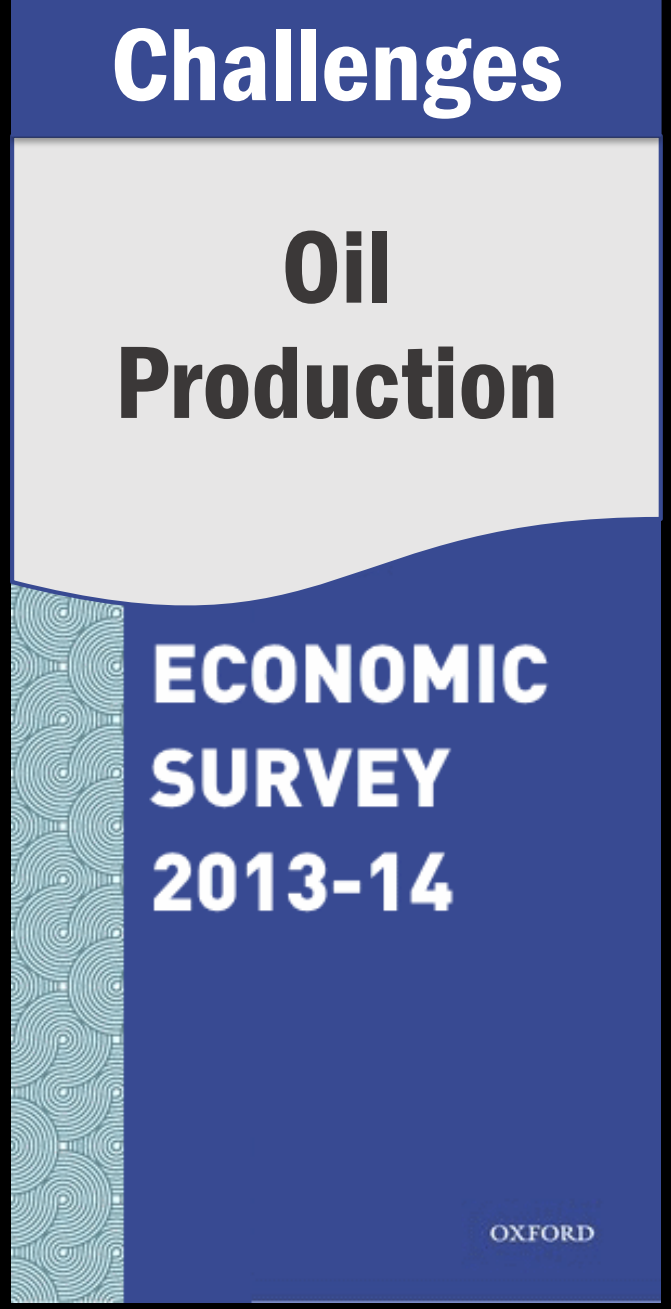
Challenges

Oil Production

1. Oil from ageing fields, except KG & RJ
2. Digboi refinery = 1st oil well of India. ~Lowest output in world.
3. Refinery -not upgraded, shortage of man, crude oil, finance
4. Challenges: environmental issues, bandhs/blockades
5. OVL get more assets abroad



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Natural gas

- Sedimentary rocks in which hydrocarbons are trapped often hold gas, sometimes in association with crude oil and sometimes alone.
- It consists mostly of methane, which is lighter than air and toxic. It therefore requires airtight tanks for storage and similarly leak-proof pipes or trucks for transport, which raise its capital costs.
- Natural gas occupies 4 times the space of a gasoline-equivalent energy

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Natural Gas in India

- In India natural gas found along with Oil reserve
- India does not have exclusive natural gas reserve
- Not enough natural gas reserve
- Sometimes, Natural gas re-injected into the oilfield to maintain pressure which forces oil up to the surface

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Sources of Natural gas for India

Production in India	Import from other countries
Bombay High	Qatar (>80%)
Gujarat	Egypt
Assam	Oman
KG basin	Australia
Kaveri Basin	Saudi Arabia

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LNG Terminals

- Natural gas is liquefied (LNG) then transported to long distances
- Liquefaction plant at exporting country and re-gasification plant at importing country
- India 5 existing such LNG terminals:

1) Dahej by Petronet
2) Hazira by Shell
3) Kochi by Petronet

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LNG Terminals

4) Ratnagiri Gas and Power Pvt. Ltd., (Dabhol) MH by GAIL + NTPC

Terminals at Dabhol and Kochi – partially operational

Absence of pipeline- from source to consumers

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5th LNG terminal at Kakinada, AP

2 major LNG terminals under process

1) By a consortium of Shell, Reliance and the KSPL (Kakinada Sea port ltd.)
2) By AP Gas Distribution Corporation Limited (a joint venture of Gail Gas and AP Gas Infrastructure Corporation) and KSPL

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5th LNG terminal at Kakinada, AP

3) floating storage and Regasification unit (Shell, Andhra Pradesh Gas Distribution Corporation (APGDC), GDF Suez)

1st of such kind in India

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Proposed new LNG terminals:

1) Mundra (GJ) by GSPC – gas from Australia and Egypt
2) Pipavav port (GJ)
3) Ennor (TN) by OIL
4) Manglore (KN) by ONGC-BPCL
5) Paradip (Odisha) GAIL dropped plan for floating terminal at Kakinada
6) Gangawaram, AP, Petronet

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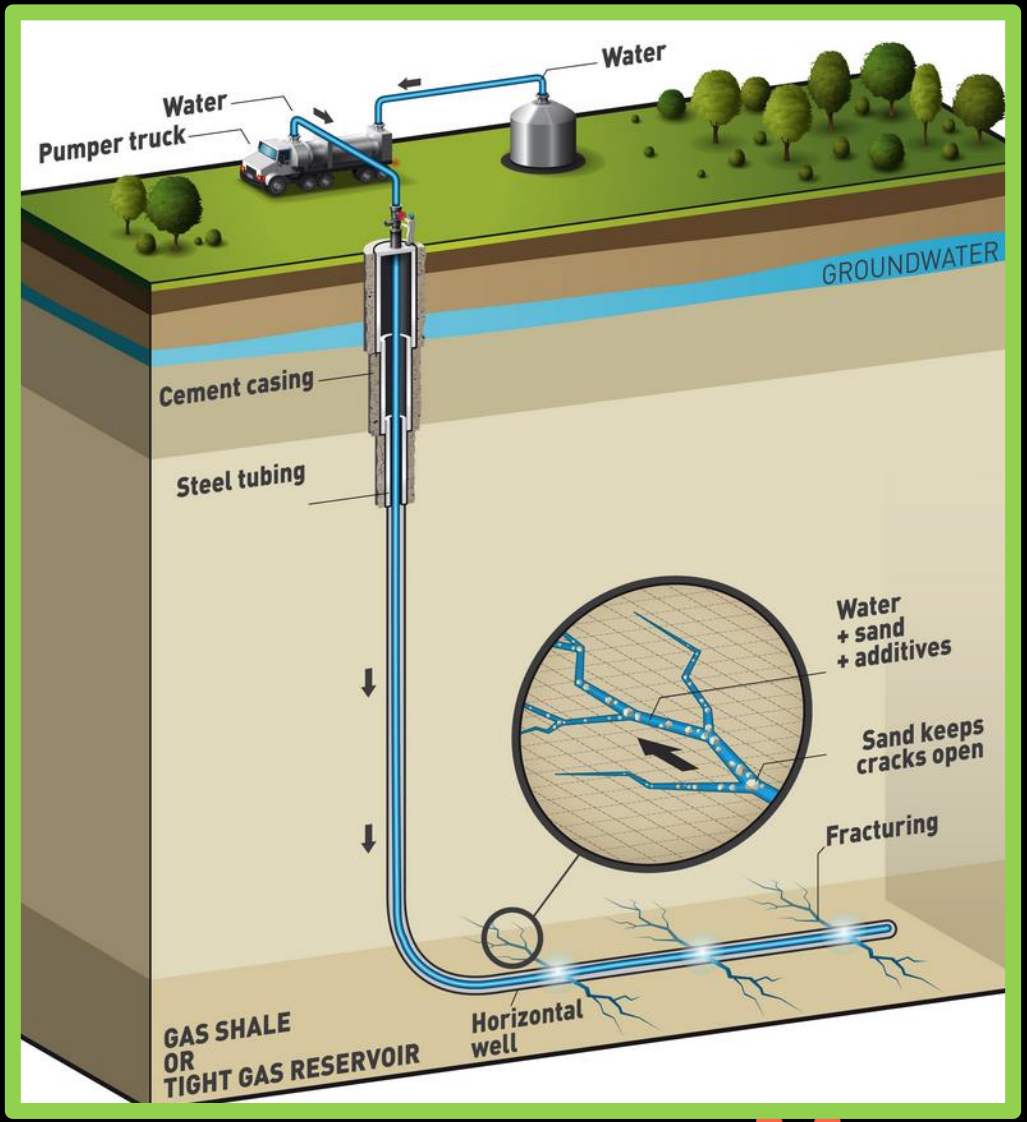
Basin centered Gas



- one type of unconventional gas
- also called “tight gas sands”
- In the deeper parts of sedimentary basin.

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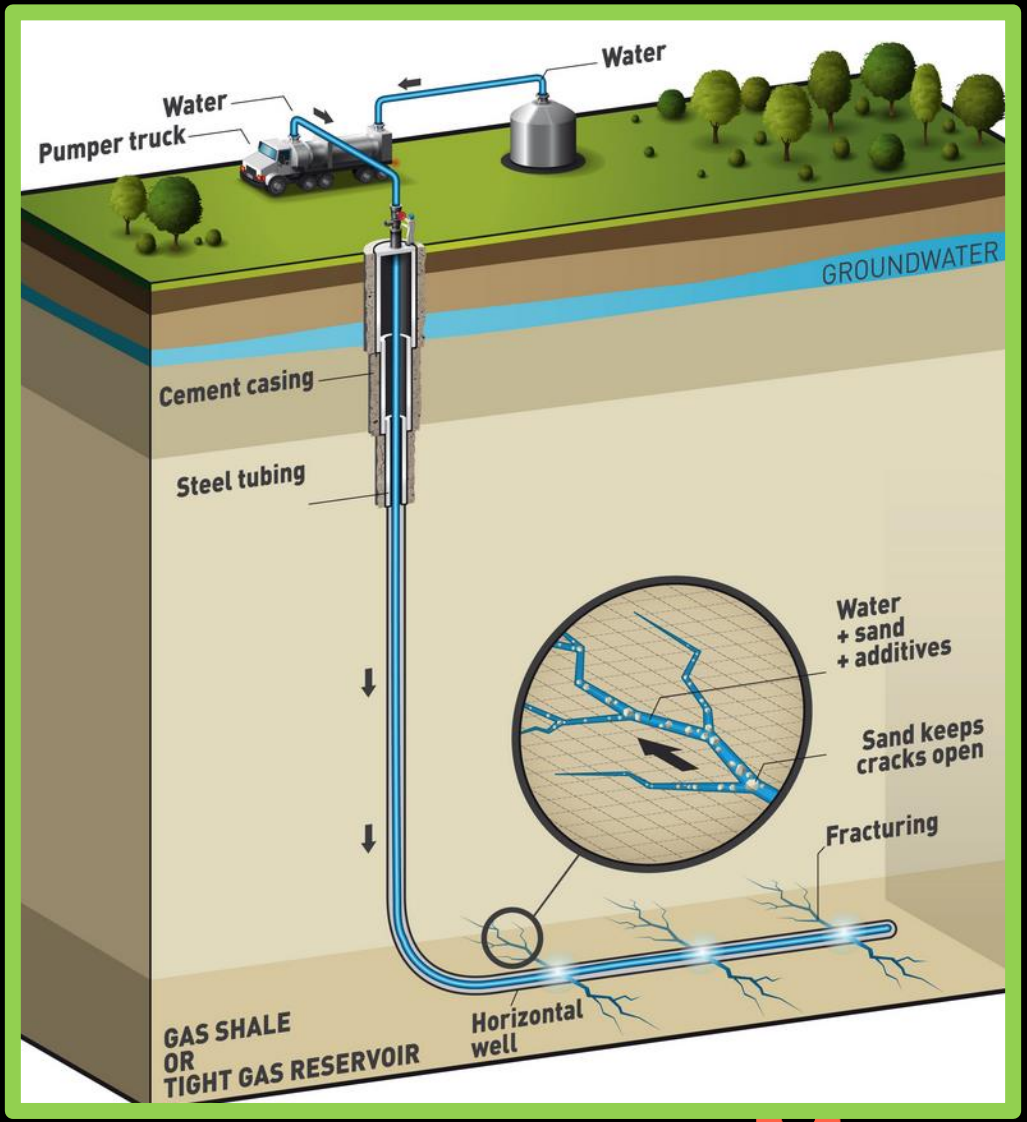
Basin centered Gas



- Their Reservoir has no Nature fractures, have to use hydraulic fracturing technique to extract gas.
- technique employed to extract gas from reservoirs without natural fractures

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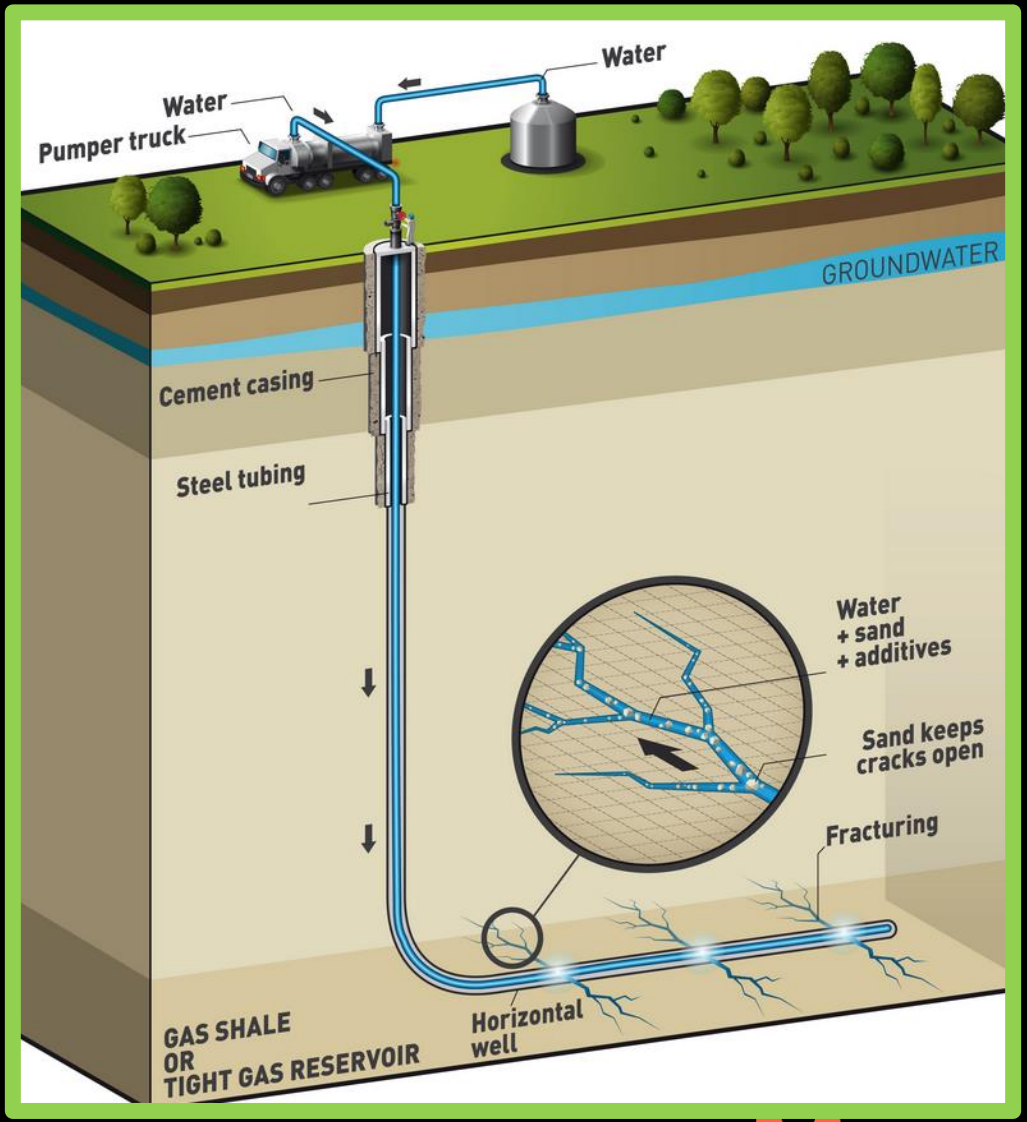
Basin centered Gas



- drilling=> special fluid pumped
- fluid contains water + sand + chemicals
- This pumping creates fractures in the basin.
- Gas migrates into the well.

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Basin centered Gas

- 15% of US Gas production via Hydraulic fracturing technique
- India doing trials since 2010

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- Gas requires an expensive transport network in the form of pipelines
- Self study: Domestic pipelines network: chap energy resources; D.R. Khullar

Self Study

Pipelines in India

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Cross-border Pipelines

- TAPI
- IPI
- Russia-India

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TAPI pipeline: detail

Pipeline project	Supply of oil	period	Operational from	distribution
TAPI	90 ml std cm/day	for 30 years	2018	India= 38 Pakistan =38 Afghanistan = 14

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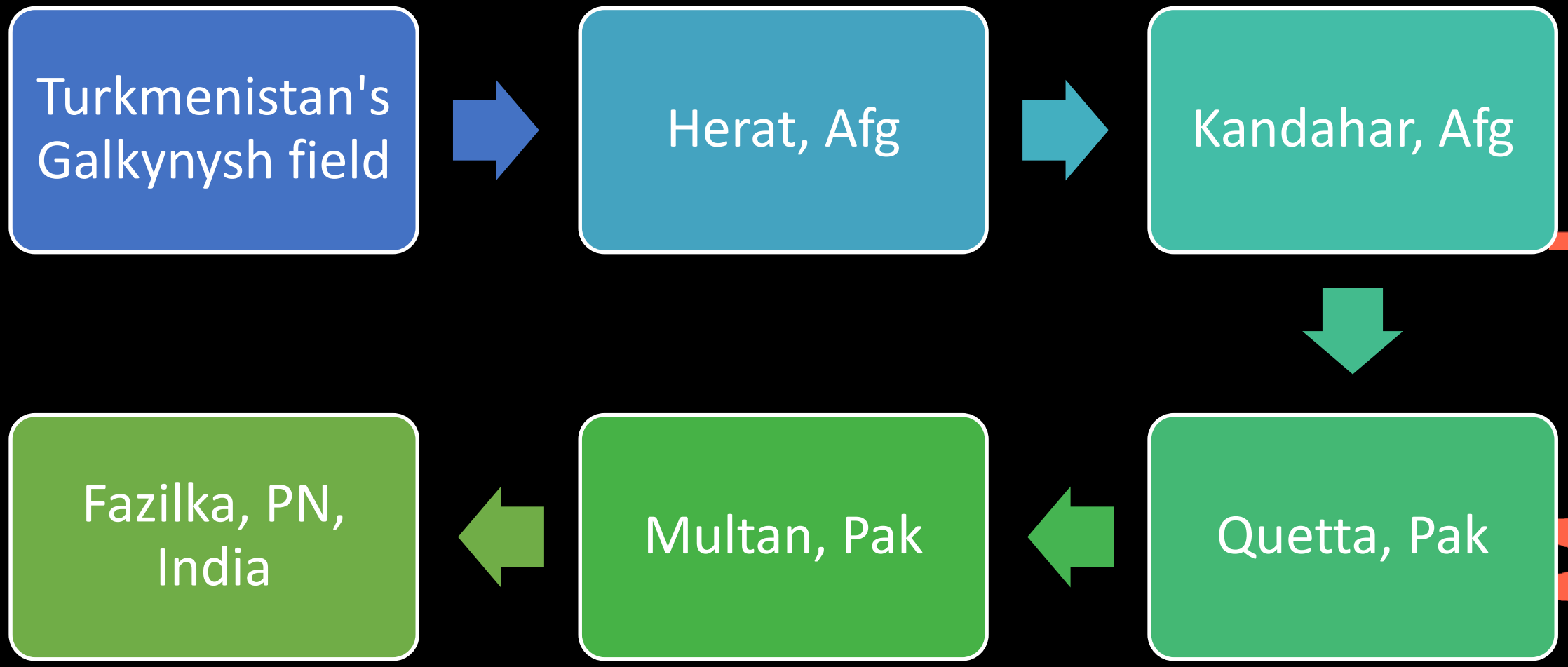
Route of TAPI



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TAPI pipeline: challenges

- Passing from politically fragile region – Afghanistan and Pakistan -Security issues
- TAPI nations have not been able to get an international firm to head a consortium, which will lay and operate the pipeline.
- International firm to construct cross-border pipeline and operate in hostile condition of Afghan-pak region

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TAPI pipeline : challenges

- French co. Total SA had showed interest in leading a consortium of national oil companies of the 4 nations.
- However, it backed off after Turkmenistan refused to accept its condition of a stake in the gas field that will feed the pipeline
- Turkmenistan -law does not provide for giving foreign firms an equity stake in gas field, without which western energy giants will not be interested to take the risk.

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TAPI and IPI

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IPI Pipeline: challenges

The pipeline faced setbacks due to sanctions put by USA on Iran

- The agreement between Pak and Iran in 2009
- Iran has constructed the pipeline on its part, while Pakistan could not due to lack of funds
- Iran has given loan to Pakistan to build the pipeline on its part

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IPI Pipeline: challenges

- But in 2013, Iran cancelled the $500 bl loan to Pakistan –sanctions on Iran, due to which Iran itself facing resource crunch
- Pakistan will have to pay penalty - $ 3 ml/day if pipeline not constructed on time
- Saudi Arabia reservation against Iran-Pakistan pipeline

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India and IPI

- Pipeline passes through disturbed region of Pakistan- Baluchistan.
- So, during hostile situation Pakistan can cut-off the supply to India, even Pakistan's, militant forces can also disrupt the supply anytime.

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Deep sea pipeline: Iran-India

- India is planning of an under-sea gas pipeline from Iran to Indian coast
- India have started talks with Iran and Oman for under-sea gas pipeline
- The pipeline is technically and economically viable.
- The construction of pipeline can be started only after sanctions against Iran are lifted
- The pipeline expected to carry 31 million cubic meter of gas per day

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Russia-India pipeline

- Natural gas pipeline
- 2 proposed routes:

1) from Russia's southern border to India via the Himalayas
2) from Russia - Astrakhan - Khazakstan — Uzbekistan and then along the Turkmenistan-Afghanistan-Pakistan route to India (TAPI gas pipeline).

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Energy security

- India = 2.4% of annual world energy production but consumes 3.4% of annual world energy production
- Future concern:
- Growing population, urbanization, Make in India campaign – energy security is necessary

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Energy security: domestic challenges

- Need Full exploration of India's sedimentary basin
- Need to Build Strategic reserve
- Problem of high transmission and distribution losses
- Untapped renewable resources (30% of its potential)

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Energy security: global challenges

- India relied on import for 75-80% of its demand
- West Asia – political fragile region
- Need diversification of sources
- Need acquisition of foreign oil fields by state oil companies

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- ONGC Videsh Ltd
- from Sudan, Vietnam, Venezuela, Brazil, Russia, Syria, South Sudan, and Colombia
- Geopolitical problem: S. Sudan, Syria = Lower production

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Energy security: global challenges

- Shale gas discovery in USA – slow down the research in renewable energy
- Developed nations not willing to transfer green technology or advanced efficient technology to other nations

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Kelkar panel on energy security (2013)

- Enhance domestic oil and gas production
- Lower down the import dependence by 2030
- Devise roadmap for deregulation of retail oil and gas
- Utilize oil industry development board cess

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Strategic reserve of crude oil

- 2004 – Mangaluru and Padur in East coast and Vishakhapatnam in west coast
- 5 ml ton
- To cover 2 weeks' requirements
- SPV- Indian Strategic Petroleum Reserve (ISPR) formed

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Strategic Reserve of crude oil

- 2015:
- Vishakhapatnam (AP)
- Bikaner (RJ)
- Rajkot (GJ)
- Padur (KN)
- Chandikhole (Odisha)
- Total 12.5 ml ton
- Cover 90 days requirement

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Innovation in Strategic reserve of Crude oil

- Iran – large crude carriers = floating storage (20-25 ml ton on tankers)
- Storage capacity can be leased out to few of oil producing nations
- Special cess or small import duty can be levied to fund the storage
- US strategic reserve = 727 ml ton
- China strategic reserve = 170 ml ton

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Question

UPSC

Prelims 2013

Q. Consider the following statements

1. Natural gas occur in Gondwana beds
2. Mica occurs in abundance in Kodarma
3. Dharwad is famous for petroleum

Which of the above statements is/are correct?

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a) 1 and 2
b) 2 only
c) 2 and 3
d) None

Ans.
b)

Question

UPSC

Prelims
2013

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